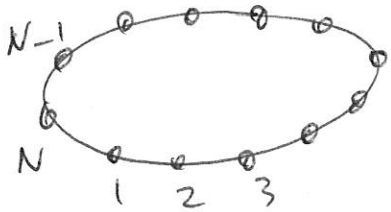
Monoatomic chain

$$H = \sum_{j=1}^{N} \frac{p_j^2}{2m} + \frac{1}{2} m\omega_0^2 \sum_{j=1}^{N} (u_j - u_{j+1})^2 \qquad \left(\begin{matrix}\text{PBC} \\ u_{N+1} = u_1\end{matrix}\right)$$



position of <u>j-th</u> atom:

$R_j = na\,j + u_j$

(equilibrium) displacement

p_j : conjugate momentum

$$\dot{p}_j = -\frac{\partial H}{\partial q_j} \Rightarrow \dot{p}_j = -\frac{\partial H}{\partial u_j} = -\frac{1}{2} m\omega_0^2 \frac{\partial}{\partial u_j}\left\{(u_j - u_{j+1})^2 + (u_{j-1} - u_j)^2\right\} =$$

$$= -m\omega_0^2 \left(u_j - u_{j+1} - (u_{j-1} - u_j)\right)$$

$$\Rightarrow \dot{p}_j = -m\omega_0^2 \left(2u_j - u_{j+1} - u_{j-1}\right)$$

$$\dot{u}_j = \frac{\partial H}{\partial p_j} = \frac{p_j}{m}$$

$$\Rightarrow \ddot{u}_j = \frac{1}{m}\dot{p}_j = -\omega_0^2 (2u_j - u_{j-1} - u_{j+1}) \quad \text{(Newton's law)}$$

Let $u_n = e^{iqna - i\omega_q t} u_q$ (plane wave solution)

$$-\omega^2 e^{iqna - i\omega_q t} u_q = -\omega_0^2 e^{iqna - i\omega_q t} u_q \left(2 - e^{-iqa} - e^{iqa}\right)$$

$$\omega_q^2 = \omega_0^2\, 2(1 - \cos qa) = \omega_0^2 \cdot 4 \sin^2\left(\frac{qa}{2}\right)$$

$$\omega_q = 2\omega_0 \left| \sin \frac{qa}{2} \right|$$

What are q's? Nothing should change if we shift all coordinates by la (l = integer)

PBC tells us that $u_{N+1} = u_1$

$$e^{iq(N+1)a} = e^{iqa}$$

$$e^{iqNa} = 1$$

$$qNa = 2\pi l \quad (l = \text{Int})$$

$$q = q_l = \frac{2\pi l}{Na}$$

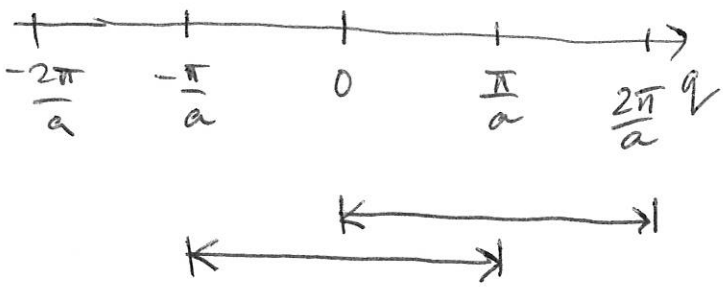
$$qa = q_l a = \frac{2\pi l}{N} \qquad \underbrace{l = 0, 1, 2, \ldots, N-2, N-1}$$

but for $l = N$

$$e^{iq_l a} = e^{i2\pi} = e^{i0} = 1$$

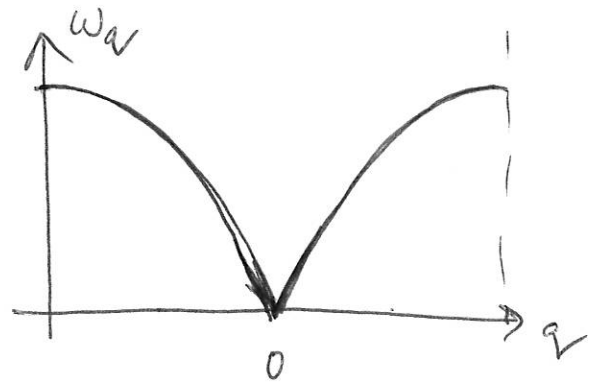
$\Downarrow$

q_l takes on N independent values



We <u>choose</u> $-\frac{\pi}{a} \le q \le \frac{\pi}{a}$

(Brillouin zone)



$q \to 0$

$$\omega_q \to 2\omega_0 \left(\frac{qa}{2}\right) = \omega_0 a q$$

<u>acoustic ~~branch~~ dispersion</u>

Laplace formula $c^2 = \frac{\partial P}{\partial \rho}$

Chain of length $L \Rightarrow L + dL$ causes restoring force

$$dF = -m\omega_0^2 \frac{dL}{N} \Rightarrow dP = -m\omega_0^2 \frac{dL}{N} \text{ too}$$

m 1d

Mass Density changes by $d\rho = -\frac{m}{a^2} da = -\frac{m\, dL}{Na^2}$

$$\rho_0 = \frac{mN}{Na} = \frac{m}{a} \rightarrow \frac{m}{a+da} = \frac{m}{a}\left(1 - \frac{da}{a}\right)$$

$$L = Na \rightarrow L + dL = N(a+da)$$

$$da = \frac{L+dL}{N} - a = \frac{dL}{N}$$

$$c^2 = \frac{-m\omega_0^2 \frac{dL}{N}}{\frac{-m\, dL}{Na^2}} = \omega_0^2 \frac{Na^2}{N} = \omega_0^2 \frac{Na^2}{N} = \omega_0^2 a^2$$

$$\Rightarrow c = \omega_0 a$$

$$\omega_q = c|q|$$

That is, change spacing $a \rightarrow a + da$

$$\Rightarrow L \rightarrow Na + Nda \Rightarrow dL = Nda$$

$$dF = -m\omega_0^2 da = -m\omega_0^2 \frac{dL}{N}$$

$$x(t) = x_0 \cos\omega t + \frac{p_0}{m\omega}\sin\omega t$$

$$p(t) = p_0 \cos\omega t - m\omega x_0 \sin\omega t$$

$$H(t) = \frac{1}{2m}\left(p_0\cos\omega t - m\omega x_0 \sin\omega t\right)^2 + \frac{1}{2}m\omega^2\left(x_0\cos\omega t + \frac{p_0}{m\omega}\sin\omega t\right)^2$$

$$= \left(\frac{p_0^2}{2m} + \frac{1}{2}m\omega^2 x_0^2\right)\cos^2\omega t + \left(\frac{(m\omega x_0)^2}{2m} + \frac{1}{2}m\omega^2\left(\frac{p_0}{m\omega}\right)^2\right)\sin^2\omega t$$

$$+ \cos\omega t \sin\omega t \left[\frac{1}{2m}(-2p_0 m\omega x_0) + \frac{1}{2}m\omega^2 \, 2x_0 \frac{p_0}{m\omega}\right]$$

$$-\omega p_0 x_0 \qquad\qquad \omega x_0 p_0$$

$$= \frac{p_0^2}{2m} + \frac{1}{2}m\omega^2 x_0^2$$

<u>Ladder operators</u> : recall that $[\hat{x}, \hat{p}] = i\hbar$

Position representation $\hat{p} = -i\hbar\frac{\partial}{\partial x}$

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$$

Time-independent $\hat{H}\psi = E\psi$

$$\psi \Rightarrow \psi e^{-iEt/\hbar}$$

$$\begin{cases} a = \sqrt{\dfrac{m\omega}{2\hbar}}\left(\hat{x} + \dfrac{i}{m\omega}\hat{p}\right) \\ a^+ = \sqrt{\dfrac{m\omega}{2\hbar}}\left(\hat{x} - \dfrac{i}{m\omega}\hat{p}\right) \end{cases}$$

(non-Hermitian! $a \neq a^+$)

$$[a, a^+] = \frac{m\omega}{2\hbar}\left[\hat{x} + \frac{i}{m\omega}\hat{p}, \hat{x} - \frac{i}{m\omega}\hat{p}\right]$$

$$= \frac{m\omega}{2\hbar}\left\{\left(-\frac{i}{m\omega}\right)\underbrace{[\hat{x}, \hat{p}]}_{i\hbar} + \frac{i}{m\omega}\underbrace{[\hat{p}, \hat{x}]}_{-i\hbar}\right\} = 1$$

~~III~~ = <u>Back</u> $\begin{cases} \hat{x} = \sqrt{\dfrac{\hbar}{2m\omega}}\,(a + a^+) \\ \hat{p} = i\sqrt{\dfrac{m\omega\hbar}{2}}\,(a^+ - a) \end{cases}$

$$H = \frac{1}{2m}\left(-\frac{m\omega\hbar}{2}\right)(a^+ - a)(a^+ - a) + \frac{1}{2}m\omega^2 \frac{\hbar}{2m\omega}(a + a^+)\cdot(a + a^+)$$

$$= -\frac{\omega\hbar}{4}\left(\underline{\underline{a^+}}a^+ - a^+a - aa^+ \underline{+aa}\right)$$

$$+ \frac{\omega\hbar}{4}\left(\underline{aa} + aa^+ + a^+a + \underline{\underline{a^+}}a^+\right)$$

$$= \frac{\omega\hbar}{2}\left(aa^+ + a^+a\right) = \frac{1}{2}\hbar\omega\left(2a^+a + 1\right) = \hbar\omega\left(a^+a + \tfrac{1}{2}\right)$$

$$[a, a^+] = 1 = aa^+ - a^+a \Rightarrow aa^+ = 1 + a^+a$$

<u>Number operator</u> $\hat{N} = a^+a$ <u>determines</u> energy of the state!

$$[N, a^+] = a^+; \quad [N, a] = -a$$

Eigenstates of $\hat{H}_{SHO} = \hbar\omega\left(\hat{N} + \frac{1}{2}\right) \longrightarrow |n\rangle$

$$E_n = \hbar\omega\left(n + \tfrac{1}{2}\right) \longrightarrow \text{state with } \underline{\underline{n}} \text{ quanta}$$

where $\hat{N}|n\rangle = n|n\rangle$

$$\hat{N}\,a^+|n\rangle = \left(a^+N + [N, a^+]\right)|n\rangle = \left(a^+\hat{N} + a^+\right)|n\rangle$$

$$= (n+1)\,a^+|n\rangle$$

And $\hat{N}\,a|n\rangle = (n-1)\,a|n\rangle$

$$\hat{N}\,\hat{a}\,|n\rangle = \left(\underbrace{[\hat{N},\hat{a}]}_{-\hat{a}} + \hat{a}\,\hat{N}\right)|n\rangle =$$

$$= \left(-\hat{a} + \hat{a}\,n\right)|n\rangle = (n-1)\,a\,|n\rangle$$

$$\Rightarrow \quad a\,|n\rangle = c\,|n-1\rangle$$

$$c^2\langle n-1|n-\rangle = c^2$$

$$\langle n|a^+ a|n\rangle = n\,\langle n|n\rangle = n$$

$$\Rightarrow a|n\rangle = \sqrt{n}\,|n-1\rangle$$

Hence the ladder is not infinite (going down)

there is a state $|0\rangle$ such that $\underline{a|0\rangle = 0}$.

All other states can be built from it.

~~Since $N = \langle n | \hat{N} | n \rangle = \langle n | a^+ a | n \rangle = \underbrace{(a|n\rangle)^+ (a|n\rangle)}_{\langle \psi_n | \psi_n \rangle} \geq 0$~~

~~$\Rightarrow$ there is minimal $n = 0$~~

All states are constracted from it:

$$|n\rangle = \frac{(a^+)^n}{\sqrt{n!}} |0\rangle$$

$$\langle n | a a^+ | n \rangle = \langle n | \left(a^+ a + [a, a^+] \right) | n \rangle$$

$$= \langle n | n + 1 | n \rangle = n + 1$$

$$\Rightarrow \quad a^+ |n\rangle = \sqrt{n+1}\, |n+1\rangle \quad \text{(up to phase)}$$

$$|n+1\rangle = \frac{a^+}{\sqrt{n+1}} |n\rangle$$

$$|n\rangle = \frac{a^+}{\sqrt{n}} |n-1\rangle$$

$$|n-1\rangle = \frac{a^+}{\sqrt{n-1}} |n-2\rangle$$

$$\Rightarrow |n\rangle = \frac{a^+ a^+ a^+ \ldots}{\sqrt{n \cdot (n-1)(n-2) \ldots}} |0\rangle = \frac{(a^+)^n}{\sqrt{n!}} |0\rangle$$

Next: linear chain

- via equs of motion
- via a_k^+, a_k operators

"Complete square"

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2 = \hbar\omega\left(\frac{p^2}{2m\omega\hbar} + \frac{m\omega}{2\hbar} x^2\right)$$

$$= \hbar\omega \left\{ \left(\sqrt{\frac{m\omega}{2\hbar}}\, x - \frac{i}{\sqrt{2m\omega\hbar}}\, p\right)\left(\sqrt{\frac{m\omega}{2\hbar}}\, x + \frac{i}{\sqrt{2m\omega\hbar}}\, p\right) + \frac{1}{2} \right\}$$

but $[x, p] = i\hbar$

$$\frac{m\omega}{2\hbar} x^2 + \frac{p^2}{2m\omega\hbar} + \sqrt{\frac{m\omega}{2\hbar}}\, \frac{i}{\sqrt{2m\omega\hbar}} \underbrace{(xp - px)}_{i\hbar}$$

$$\Rightarrow H = \hbar\omega\left(a^+ a + \frac{1}{2}\right).$$

This is the simplest way to relate H_{SHO} to $\hat{a}^+\hat{a}$.

Do Fourier decomposition earlier!

$$\begin{cases} u_j = \frac{1}{\sqrt{N}} \sum_{k \in BZ} e^{ikaj} u_k \\ p_j = \frac{1}{\sqrt{N}} \sum_k e^{ikaj} p_k \end{cases} \qquad \begin{cases} u_k = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{-ikaj} u_j \\ p_k = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{-ikaj} p_j \end{cases}$$

$$[u_j, p_{j'}] = i\hbar \, \delta_{jj'}$$

Note that $u_{-k} = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{ikaj} u_j$

real (Hermitian)

$$\Rightarrow \begin{cases} u_{-k} = u_k^+ \\ p_{-k} = p_k^+ \end{cases}$$

$$[u_k, p_{k'}] = \frac{1}{N} \sum_{j=1}^{N} \sum_{j'=1}^{N} e^{-ikaj} e^{-ik'aj'} \underbrace{[u_j, p_{j'}]}_{i\hbar \, \delta_{jj'}}$$

$$= \frac{1}{N} \underbrace{\sum_{j=1}^{N} e^{iaj(k+k')}}_{N \delta_{k,-k'}} (i\hbar) = i\hbar \, \delta_{k,-k'}$$

$$\frac{1}{2m}\sum_{j=1}^{N} p_j^2 : \sum_{j=1}^{N} \underbrace{\frac{1}{N}\sum_{k,k'} e^{ikaj+ik'aj}}_{\delta_{k,-k'}} P_k P_{k'} =$$

$$= \sum_{k\in BZ} P_k P_{-k}$$

$$\sum_{j=1}^{N}\left(u_j - u_{j+1}\right)^2 = \frac{1}{N}\sum_j \sum_{kk'} \left(e^{ikaj} - e^{ika(j+1)}\right) u_k \cdot$$

$$\cdot \left(e^{ik'aj} - e^{ik'a(j+1)}\right) u_{k'} = \frac{1}{N}\sum_j \sum_{kk'} e^{i(k+k')aj}$$

$$\left(1-e^{ika}\right)\left(1-e^{ik'a}\right) u_k u_{k'} =$$

$$= \sum_{k\in BZ}\left(1-e^{ika}\right)\left(1-e^{-ika}\right) u_k u_{-k} =$$

$$= \sum_{k\in BZ}\left(1+1-2\cos ka\right) u_k u_{-k} = \sum_k 4\sin^2\left(\frac{ka}{2}\right) u_k u_k$$

$$\Rightarrow H = \sum_{k\in BZ} \frac{1}{2m} P_k P_{-k} + \frac{1}{2} m\omega_0^2 \cdot 4\sin^2\left(\frac{ka}{2}\right) u_k u_{-k}$$

$$= \cancel{\frac{1}{2m}} \sum_{k\in BZ} \frac{1}{2m} P_k P_{-k} + \frac{1}{2} m \Omega_k^2 u_k u_{-k}$$

$$\Omega_k = 2\omega_0 \left|\sin\frac{ka}{2}\right|$$

N harmonic oscillators with different frequencies

Now play SHO game:

$$a_k = \sqrt{\frac{m\Omega_k}{2\hbar}}\, u_k + \frac{i}{\sqrt{2\hbar m\Omega_k}}\, p_k$$

$$a_k^+ = \sqrt{\frac{m\Omega_k}{2\hbar}}\, \underset{\substack{\downarrow \\ u_k^+}}{u_{-k}} - \frac{i}{\sqrt{2\hbar m\Omega_k}}\, p_{-k}$$

$$[a_k, a^+_{+k'}] = \sqrt{\frac{m\Omega_k}{2\hbar}}\, \frac{1}{\sqrt{2\hbar m\Omega_k}} \Big(-i \underbrace{[u_k, p_{-k'}]}_{i\hbar\delta_{k,+k'}}$$

$$+ i \underbrace{[p_k, u_{-k'}]}_{-i\hbar\delta_{k,+k'}} \Big) = \frac{1}{2\hbar}\, \hbar\delta_{k,-k'}(1+1) = \delta_{k,+k'}$$

Thus $a_k^+ a_k = \frac{1}{2\hbar m\Omega_k} p_{-k}p_k + \frac{m\Omega_k}{2\hbar} u_{-k}u_k$

$$+ \left(-\frac{i}{2\hbar} p_{-k}u_k + \frac{i}{2\hbar} u_{-k}p_k \right) \longleftarrow \frac{i}{2\hbar}(i\hbar)$$

$$[u_k, p_{k'}] = i\hbar\delta_{k,-k'} \Rightarrow u_k p_k - p_{-k}u_k = i\hbar$$

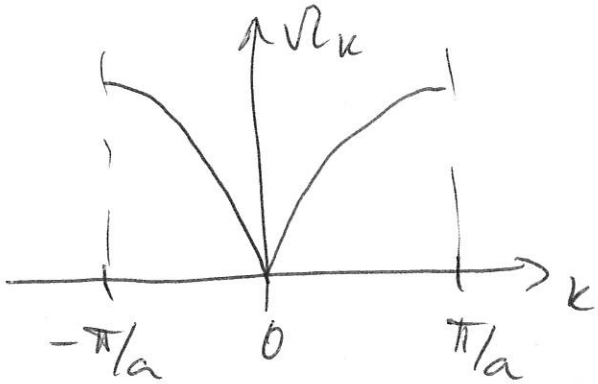
$$\hbar\Omega_k a_k^+ a_k = \frac{1}{2m} p_{-k}p_k + \frac{1}{2} m\Omega_k u_{-k}u_k - \frac{1}{2}\hbar\Omega_k$$

Thus $H = \sum_{k \in BZ} \hbar \Omega_k \left(a_k^+ a_k + \frac{1}{2} \right)$

$$= \sum_{k \in BZ} \frac{1}{2m} P_{-k} P_k + \frac{1}{2} m \Omega_k^2 u_{-k} u_k$$

$$= \sum_{j=1}^{N} \frac{1}{2m} P_j^2 + \frac{1}{2} m \omega_0 (u_j - u_{j+1})^2 .$$

- a_k^+ (a_k) create (annihilate) phonons with energy $\hbar \Omega_k$
- phonons = quanta of lattice vibrations
- key to many phenomena including <u>superconductivity</u>



$\Omega_k \approx \omega_0 a |k|$ for $k \to 0$